

GI Hub - User Manual (v2, React/FastAPI)

General Industries - GI Hub ERP

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GI Hub - User Manual (React / FastAPI stack)

Version 2.1 - 2026-07-18 (pre-deploy final). This manual covers the NEW web application (React + FastAPI + PostgreSQL) that replaced the Streamlit app. It is organised by role - read §1-§2 first (everyone), then your role's chapter. The legacy manual for the retired Streamlit app remains at the repository root (USER_MANUAL.md) for historical reference.

Roles at a glance

Role	Level	Home page	Chapter
Store Keeper (SK)	0	Data Entry -> Issue	§3
Supervisor	1	Supervisor -> Material Requests	§7
Warehouse User	1	Warehouse	§6
Head of Department (HOD)	2	HOD -> Approvals	§4
Logistics	3	Logistics -> Procurement	§5
Admin	4	Admin -> Console	§8

Admin can open every portal (an "All areas" sidebar toggle reveals the operational ones). Every other role sees only its own pages - both in the sidebar and at the API.

1. Getting started (everyone)

1.1 Signing in

Open the site (production: <https://gi.giinventory.com>), enter your username and password. If your account has two-factor enabled you will be asked for the 6-digit code from your authenticator app. New colleagues use Request access on the login page - an admin approves the request and the login starts working.

Sessions refresh silently in the background; you stay signed in across reloads until you Sign out (top-right) or an admin revokes the session.

1.2 Installing the app on your phone (PWA)

The site is an installable Progressive Web App:

- Android/Chrome: menu -> Add to Home screen -> Install.
- iPhone/Safari: Share -> Add to Home Screen.

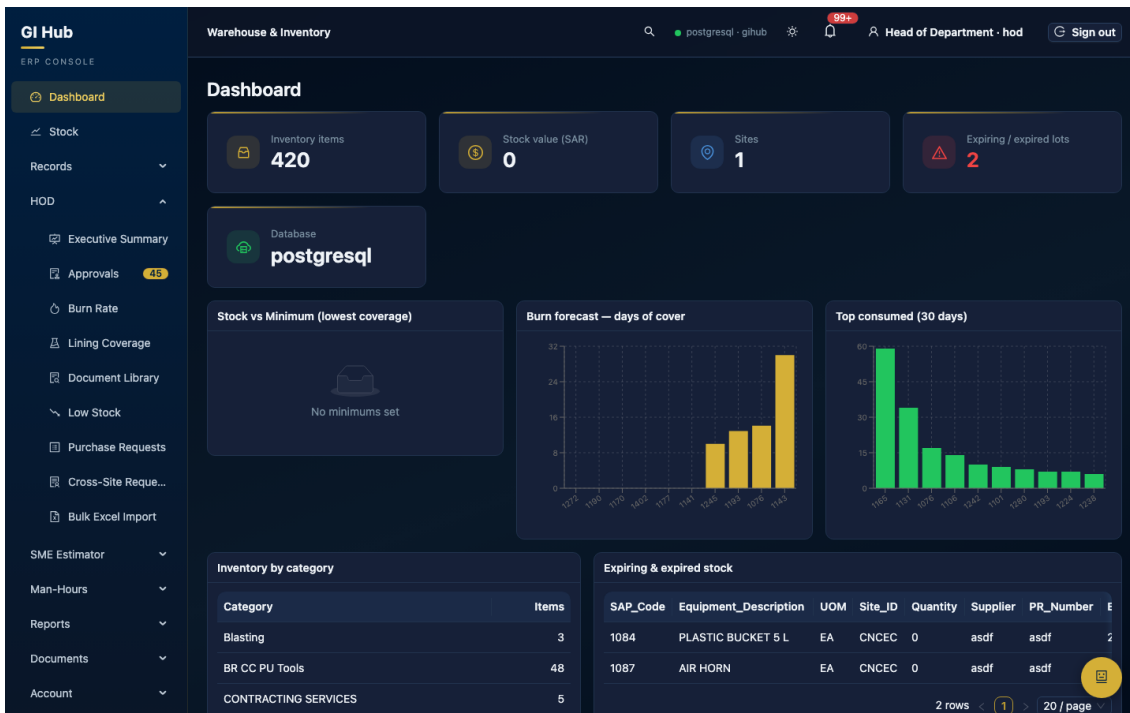
The installed app opens full-screen, keeps you signed in, and caches the read pages for offline viewing.

1.3 Working offline (Store Keepers)

Read pages (stock, records, notifications) show the last good copy when the connection drops. The four material-entry forms (Issue / Receive / Return / Adjust) keep working fully offline: a submission with no network is

saved to the device's sync queue - you'll see an amber "saved offline" toast and a cloud-sync badge in the header with the queued count. When the connection returns (or you tap the badge) the queue replays automatically; rows the server rejects are dropped and reported, so one bad row never blocks the rest. Approvals, admin and auth actions deliberately stay online-only.

1.4 Common chrome



- Health dot (header): green + postgresql - gihub = connected.
- Bell: every action that concerns you (approvals, rejections, DNS, loans, alerts) lands here; clicking a row marks it read and jumps to the right page. Significant events are ALSO WhatsApp'd to the concerned person.
- CmdK / Ctrl-K: command palette - fuzzy-search every page your role can open and jump to it.
- Theme toggle: dark <-> light, remembered per device.
- Profile -> phone number: self-service change with a dual WhatsApp OTP - a code to your OLD number authorises the change, a second code to the NEW number proves it can receive messages before anything is saved. (Admins can override a number directly.)
- Urgent / Evening toggle (on the entry forms only): "Evening" batches your notification into the 16:00 daily digest instead of an immediate WhatsApp; critical alerts always go out immediately.
- WhatsApp commands (send to the company number): STOCK <SAP> returns the live stock of an item at your site; RESET PASSWORD issues a temporary credential and signs out all your sessions.
- Tables: column headers stay frozen at the top while you scroll long lists, and quantities show clean numbers - 5, not 5.00 (real fractions keep their decimals).
- Feedback (everyone): file a [bug] bug or [feature] feature request from the

Feedback page - give it a short title and pick how urgent it is. You are notified when an admin responds; track yours in "My reports".

2. Documents, gates and the AI - concepts used everywhere

2.1 Supporting-document gate

While the admin setting `require_entry_documents` is ON (the production default), every Issue / Receive / Return submission **must** carry at least one uploaded supporting document^{**} - photograph the hand-written note or delivery note with the Photograph note button, or attach a file. Uploads are filed per site and date and appear in the HOD Document Library with inline preview; approvers see a on every pending row.

Additional gates, independent of that switch:

- MTC - receipts of materials in the Surface Shields category are hard-blocked until an MTC certificate is uploaded (missing MTCs also email Logistics automatically).
- WBS - once the HOD configures active WBS numbers for a site, every Issue/Receive at that site must carry one.
- Returns - a return needs a Return DN No. and must reference the source receipt (last 30 days; older needs an override justification that shows as a red >30d flag in HOD approvals). Quantity is capped to the source receipt.

2.2 Reading documents with the AI (OCR doc assist)

After attaching a photo on the Issue or Receive form, press Read (AI) on the attachment chip. The vision model reads the note and pre-fills what it finds - on Receive: the date, supplier and delivery-note header (DN No., vehicle, driver - these ride along into the ledger); on Issue: the material, quantity and recipient when the note has one confidently-matched row. Notes listing many materials are better handled by Data Entry -> OCR Import, which stages the whole grid at once (with a Paste tab that works even when the AI is offline). The AI never submits anything - you always review first.

2.3 Asking your data (HOD/Admin/Logistics)

The Dashboard's Ask your data card answers plain-English questions: "How many issues in the last 30 days?", "surface shields stock", ^{**}"what was received on DN 15610?". Common questions are answered instantly by built-in templates (site-scoped automatically - an HOD only ever sees their own site, whatever they type); unscoped roles (Logistics/Admin) additionally get AI-generated SQL for free-form questions, run on a read-only database login. The executed SQL is always shown for transparency. The templates search DEEP: "current stock for surface shield category items" filters to exactly that category, and naming a material family ("furan materials", "remafix") finds the items through both the ERP descriptions and the SME recipe names - the answer line shows which filters were applied.

2.4 The Hub Assistant

The floating button opens a chat assistant that knows the user manual and your role's pages. If the local AI is offline it says so gracefully - no feature above stops working (paste lanes and manual entry always exist).

3. Store Keeper (SK)

Your portal is Data Entry. Everything you stage goes to your HOD for approval before it touches the ledger.

3.1 Receive (goods in)

GI Hub
ERP CONSOLE

Warehouse & Inventory

Receive Stock
Add each goods receipt to the batch, review the list, then submit them all at once for HOD approval — auto-creates a lot when an expiry is set, and closes the linked PR once fully received (same rules as the old app).

*** Site** **Category** *** Material (SAP Code)**

Select site All Search material

*** Quantity** *** Receipt date** **Expiry date (optional)**

0 2026-07-13 Select date

Supplier **PR Number (optional)** **Lot Number (optional)**

Supplier name links + auto-closes PR auto if expiry set

Bin / Shelf (optional)

e.g. R2-S4

Remarks

Notes (optional)

+ Add to batch

Batch (0 lines) WhatsApp alerts: **Urgent** **Evening digest** **Submit batch to HOD**

@ Supporting documents (required — hand-written note / delivery note)

Attach file **Photograph note** at least one document must be attached before submitting

Material **Qty** **Supplier** **Expiry** **PR**

1. Pick the site and material - type to search, narrow by category, or press Scan and point the camera at the item's QR/barcode label.
2. Rubber items (Surface Shields) demand an MTC upload before the line can be added; the MTC number is kept with the document.
3. Quantities can be entered in the pack unit (e.g. drum) - the form shows the conversion and the base-unit result.
4. Attach the delivery note () - then Read (AI) pre-fills the DN header and date for you (§2.2).
5. Add to batch, repeat for every line on the note, then **Submit batch to HOD**. Bin/Shelf and WBS fields appear when relevant.

3.2 Issue (consumption)

Same batch flow. The form shows the FEFO lot hint (oldest expiry first) - issuing a different lot asks for a short override reason and alerts the HOD (allowed, logged, never blocked). Over-issue beyond current stock is also allow-and-log. Issued To, Work Type, Tank No feed the reports.

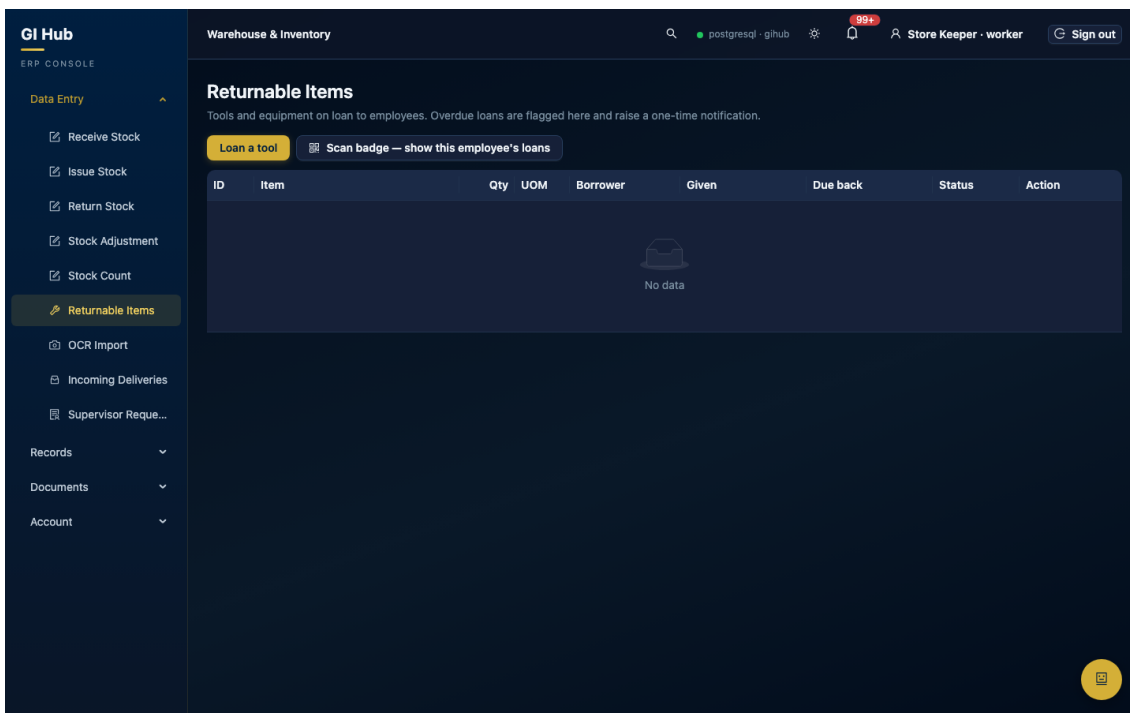
Surface Shields items (lining materials) work system-first. When your category filter (or picked item) is Surface Shields, a Lining System selector appears - pick the System Code FIRST:

1. the material picker narrows to exactly that recipe's SAP codes;
 2. the system's progress shows live - Done SQM, Pending SQM, total and unit count for your site;
 3. the batch line records the system automatically (LS <code> in Remarks) so the HOD sees which lining system consumed the material.
- Trying to add a Surface Shields material without a selected system is refused with a hint - this is deliberate: lining consumption is only meaningful against a system.

3.3 Return / Adjust / Count sheet

Returns follow §2.1's source-receipt rules. Adjust stages a correction with a reason; the Count sheet lets you enter counted quantities and stages the variances in one go.

3.4 Returnable items (tool loans)



- Loan a tool: Scan badge identifies the borrower from their employee QR badge (name + phone pre-fill); Identify tool (photo) names the tool from a photo. Set the due-back time - the borrower gets a WhatsApp with the details, and another when the tool is marked returned.
- Returns: Scan badge - show this employee's loans filters the table to the person standing in front of you; Mark returned closes the loan.
- Overdue loans turn red, notify you once, and WhatsApp the borrower.

3.5 OCR Import, incoming deliveries, SK requests

- OCR Import: photograph a full hand-written log -> review the recognised

grid (green = matched automatically, orange = pick from candidates) -> stage everything to HOD in one click.

Validate (handwritten spec) runs the full daily-consumption-form rulebook over the grid: known handwriting fixes ("Yloues"->"Gloves"...), ditto-mark (") resolution, quantity rules (2+3 sums; a blank quantity next to a product means 1), the approved substitution list for out-of-stock items, and a whole-batch stock simulation - rows that would drive stock negative are blocked with a (!!) marker ([?] = check, (!) = heads-up). TSV export then downloads the legacy 17-column tab-separated file for the old Excel/VBA sheet (blocked rows are never exported). Struck-through rows on the paper are counted but never staged.

- Incoming deliveries: warehouse shipments to your site appear here -

Receive stages the pending receipt for HOD approval.

- SK requests: supervisors' material requests (SMRs) land here - adjust quantities (0 = withdraw a line), approve to mirror into the HOD issue queue, or reject.

4. Head of Department (HOD)

4.1 Approvals

Tabs per entry type with live counts. Per row: Edit (whitelisted fields), Approve, Reject (reason required - the SK is notified either way), plus bulk approve (<=200). The button previews the supporting documents; returns older than 30 days carry a red >30d tag (hover for the justification). A banner warns when an approval would drive stock negative (allowed, logged). The DN tab is stage 2 of warehouse deliveries (\$5).

4.2 Executive Summary, Burn Rate, Low Stock

- Executive Summary: date presets, 13 sections, Download PDF (server-rendered, nothing clipped) and Excel (14 sheets). Admins get a site selector. Every Friday 17:00 the system mails admins + HODs a secure 72-hour download link via WhatsApp automatically.
- Burn rate: consumption trends per site/period.
- Low stock: items under minimum - Auto-draft PR creates a purchase request from the shortfall in one click.

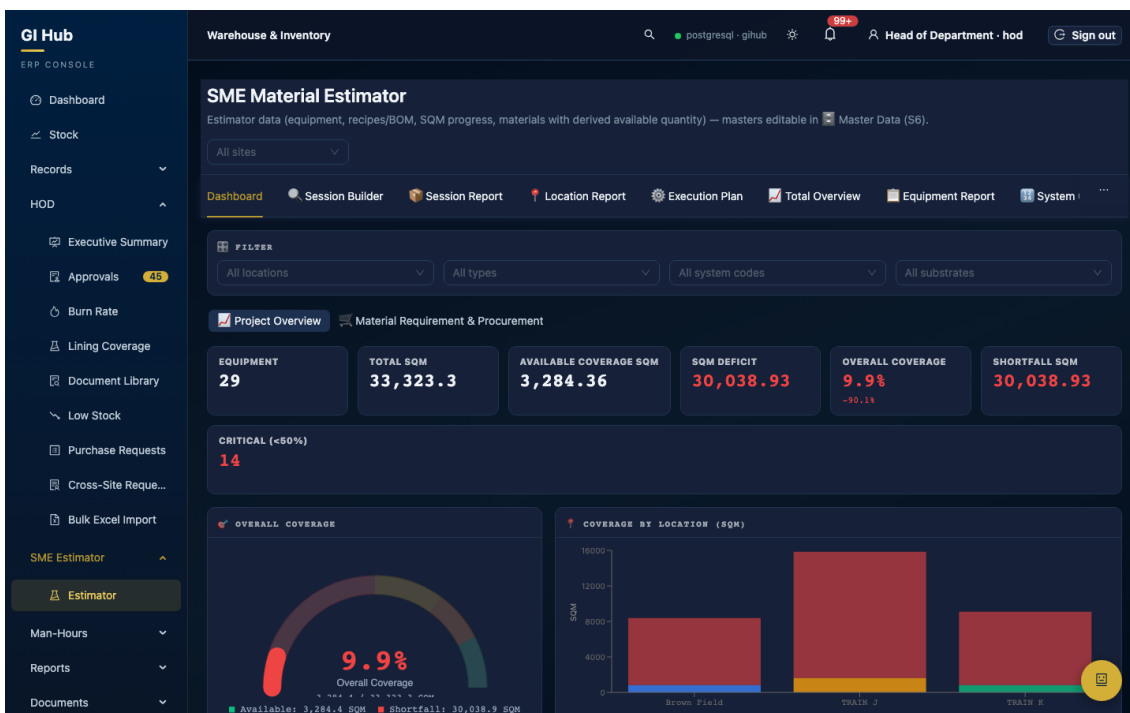
4.3 Purchase Requests

Create PRs (or import from a PDF), edit/rename lines while draft, **Submit to Logistics**, download the PR PDF at any stage. Track everything under Records -> Purchase Requests.

4.4 Document Library & site configuration

- /hod/documents: every SK upload for your site - type tabs, doc-number and date filters, inline image/PDF preview, download.
- WBS config (Site config): add/close the WBS numbers that gate entries (\$2.1).

4.5 SME Material Estimator

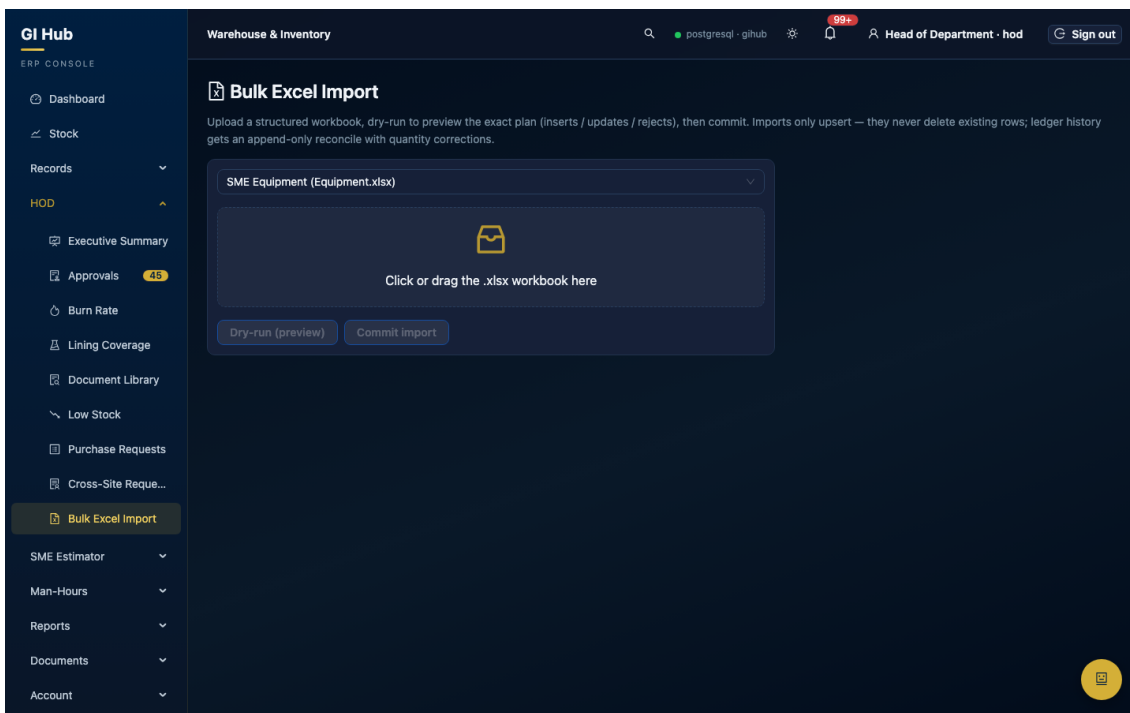


All the estimator tabs are yours: Dashboard (filters + KPI drill-downs), Session Builder (drag priorities - numbers recompute instantly), reports, Execution Plan, Total Overview, exports. Master Data (S6) edits the masters directly: Equipment (adding one also seeds its SQM-progress baseline; deleting removes it), Recipes/BOM, Materials seed (live availability stays derived from ERP movements), SQM Progress, and the Location/Type dropdowns.

The estimator and the Lining Coverage page (live ledger stock vs remaining SQM, with depletion forecasts) recompute immediately after any master-data change.

Smart Calculator (HOD/Admin, same SME page): pick a lining system and type a target SQM - you get the fully segregated component list from the recipes: per-SQM factor, exact required quantity, pack counts (from the package size), and live stock coverage per line (green or a red "short N" tag), each with a plain-language explanation row like 2.5 KG/SQM x 40 SQM = 100 KG -> 4 x 25 KG pack(s) - in stock: 150 . Use it to sanity-check a job before issuing or raising a PR. Recipes are SAP-exact since 2026-07-18 - PU systems list Comp-A/B/C/D lines separately.

4.6 Bulk Excel Import (HOD kinds)

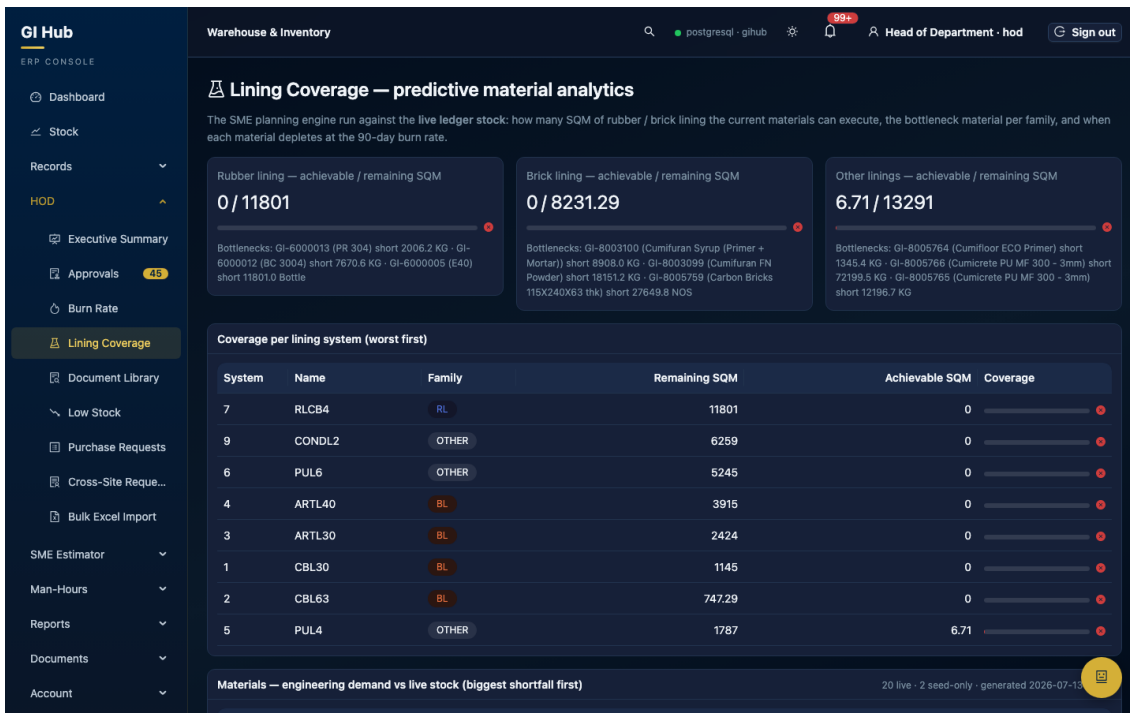


HOD -> Bulk Excel Import accepts the three SME workbooks - Equipment.xlsx, For_1_SQM.xlsx (recipes), Materials_DetailsAvailable_Qty.xlsx - with the same column layout as the originals. Dry-run first (shows exactly what would be inserted/updated/rejected and why), then Commit. Imports only upsert - nothing is ever deleted - and equipment re-imports preserve recorded Done-SQM progress. See §9 for the format rules.

4.7 Man-Hours

Employee roster, timesheet import (dry-run first), SQM distribution, estimates with auto-draft, variance + scorecard exports.

5. Logistics



- Incoming PRs: create POs from submitted PRs (PR -> in-PO), force-close with a reason (24 h undo).
- Purchase Orders: KPI hero cards filter on click; per-line force-close; Assign to warehouse starts the delivery pipeline; manual POs and inline vendor creation supported; PO PDF import available.
- DN approvals (stage 1): approve warehouse delivery notes onward to the HOD, or reject them back.
- Vendor returns: send goods back to a vendor (reopens the PO line) and close the loop.
- Reschedules / Force-closures: decide warehouse date-change requests; undo a force-close within 24 h.
- Lining Coverage: the same live RL/BL coverage page the HOD sees (§4.5).
- You also receive the automatic emails: missing MTCs and approved returns.

6. Warehouse

Assignments arrive from Logistics: Acknowledge -> Receive (partial supported) -> Prepare DN (quantities + lots per line) -> submit. The DN passes Logistics (stage 1) then the site HOD (stage 2); Ship unlocks after both approve. The destination SK receives it (§3.5), and their HOD's approval writes the ledger. Returns-from-site are recorded with a disposition (restock / scrap / return to vendor). You are pinned to your own warehouse; Logistics/Admin can pick any.

7. Supervisor

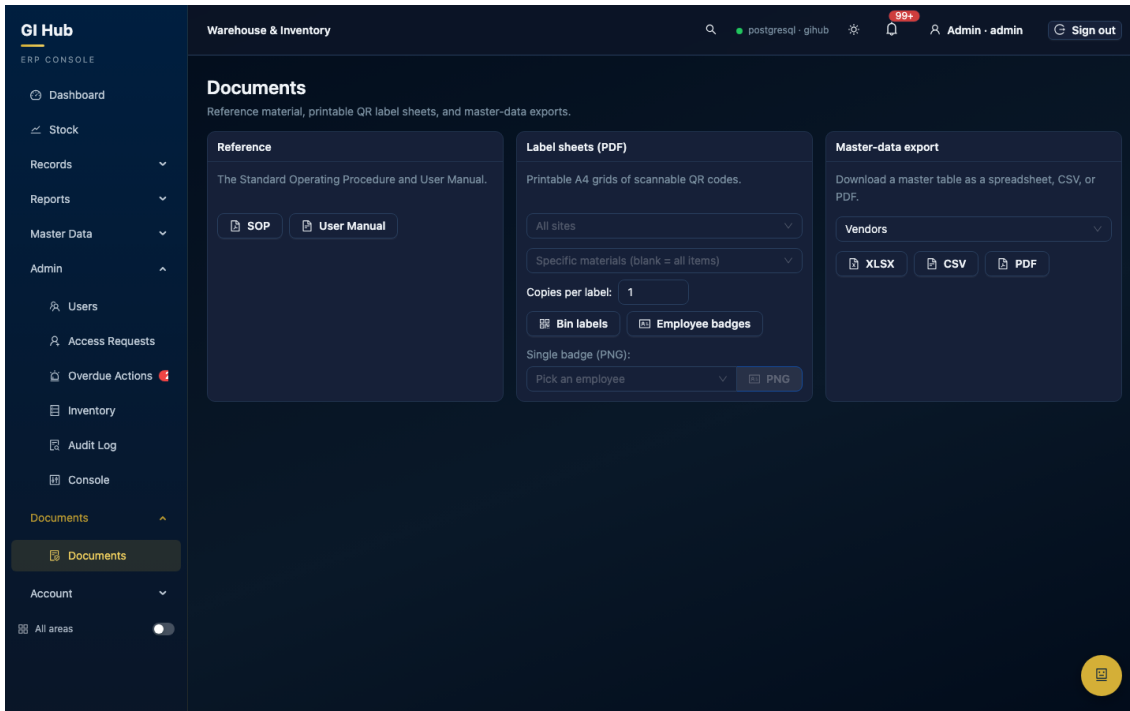
New request: pick the worker (must be an active employee at your site),

add lines - live stock hints show availability - flag old-PPE returns, and submit. The SK reviews (may adjust quantities), then the HOD approves the issue. My requests lets you cancel while still pending; the **intent vs actual** table shows requested vs finally-issued quantities.

8. Admin

Everything above (via "All areas") plus:

- Users: create (scoped roles need a site), edit, reset password/2FA, delete (self-delete and last-admin are blocked). Access requests: approve registrations into real logins.
- Console:
 - Settings: require_entry_documents (the §2.1 master switch), mtc_required_category (default Surface Shields), maintenance mode (blocks non-admin logins), thresholds.
 - WhatsApp / Email consoles: outbox status + retry.
 - Lots: quarantine <-> release -> dispose (terminal).
 - Sites, Sessions (revoke one/all), Backup (pg_dump), Oversight.
 - Feedback - the Bug Tracking Engine: every user-filed report arrives with its severity. Triage captures the analysis, the **safety constraints** (what must not break) and a rollback plan **BEFORE** any change is attempted; the submitter is notified on every status change. Prompt copies a self-contained implementation brief (the report + your triage + the project's mandatory test gates) ready to paste into a maintainer's coding session - the portal itself never changes code, so filing and triaging reports can never destabilise the system. *Export open reports* downloads a Markdown digest for batch analysis.
- Bulk Excel Import (admin kinds): the full CNCEC inventory workbook - the Inventory master sheet (upsert on SAP; categories are canonicalised so the MTC gate keeps matching) and the Ledger backfill (Receipt / Consumption / Return Log sheets; append-only with automatic duplicate matching and quantity corrections - it never deletes ledger history). Always dry-run first; §9 has the rules.
- Audit log: filterable, append-only record of every write.
- Reports: every report card in xls/pdf/csv, schedules (create / toggle / run-now), archive. The Friday executive PDF runs itself (§4.2).



- Documents: SOP + manual downloads, QR label sheets (pick specific materials and copies per label), employee badge sheets, and single-badge PNG downloads for new hires.

9. Excel bulk import - format rules

All kinds: .xlsx only, <=8 MB, header names as in the original workbooks (a title banner row above the header is fine). Dry-run first, always - the commit applies exactly the previewed plan, and every commit is audited.

Kind (who)	Sheet	Key	Behaviour
Inventory master (admin)	Inventory	SAP CODE	Upsert of master fields (description, UOM, category, opening stock, minimum). Aggregate columns (Receipt/.../Current Stock) are ignored - stock is always ledger-derived. "Surface Shield" is canonicalised to "Surface Shields". A Material_Code already owned by another SAP is only reassigned when that owner is re-mapped in the same file; otherwise the row imports without it (warning).

Kind (who)	Sheet	Key	Behaviour
Ledger backfill (admin)	Receipt Log, Consumption Log, Return Log	date + SAP + qty + DN/ref	Append-only reconcile: exact rows are matched (never duplicated); the same day+SAP+DN with a different quantity is treated as a workbook correction and updated; rows for unknown SAPs are rejected; database rows missing from the workbook are reported but never deleted.
SME Equipment (HOD)	Data Input	site + tag + code	Area-split rows are summed per (tag, code); tag-less civil areas use their Name as identity; non-numeric lining codes are skipped; re-imports re-baseline Original-SQM but preserve Done-SQM.
SME Recipes (HOD)	first sheet	code + material + SAP	With the SAP_Code column (2026-07-18 layout) a line's identity is (code, material, SAP) - PU component lines (Comp-A/B/C/D) sharing one material stay separate, and a repeated identity is a deliberate coat line whose For-1-SQM values sum. Legacy files without the SAP column keep the old first-wins rule. Comma-separated material cells still split into one line per material.
SME Materials (HOD)	first sheet	Material_Code	Multiple PO lines per code are aggregated (quantities summed, latest document date wins); with the SAP_Code column the distinct variant SAPs are recorded per material.

Columns are matched by header name - reordering or adding columns in a workbook is safe; anything unrecognised is listed in a dry-run warning instead of being silently dropped.

10. Troubleshooting quick answers

Symptom	Answer
"supporting document..." error on submit	Attach the note first (§2.1) - or the admin has the gate ON on purpose.
Receive blocked with an MTC message	The item is Surface Shields - upload the MTC certificate on the line.
"WBS Number required"	Your HOD configured active WBS numbers - pick one from the select.
Amber "saved offline" toast	No network; the entry is queued and will sync automatically (§1.3).
AI buttons say offline/unavailable	The local model is down - paste lanes and manual entry always work.
429 "try again in Ns" toast	Rate limit hit - the countdown tells you exactly when to retry.
Stock looks wrong	Stock = receipts consumption returns (approved rows only). Check pending approvals first, then Records.

Maintained with the code: update this manual in the same commit as any user-facing change. The role-based PDF slices served in-app are generated from the legacy manual until the doc pipeline is repointed post-deploy.